

2026 FALL SEMESTER
Programming for Data Science (MACSE502) Lab Record
Practice Program

Date: 20/08/2026

Week No.: 6

Program S. No. 11

Title	Interactive Bar Chart with Plotly			
Problem Statement	Create an interactive bar chart showing the population of different cities.			
Objectives	• To understand the creation of interactive data visualizations using Plotly. • To organize city population data in a Pandas DataFrame. • To create an interactive bar chart for comparing population across different cities. • To sort and analyze population data to identify the highest, lowest, and average values. • To save the interactive chart as an HTML file and the dataset as a CSV file.			
Dataset URL Link / File	City		Population (Lakhs)	
	Mumbai		204	
	Delhi		167	
	Bengaluru		136	
	Hyderabad		108	
	Chennai		115	
	Kolkata		149	
	Dataset source: Manually defined in the Jupyter Notebook for this exercise; no external dataset URL is required.			
Functions Details	Sl. No.	Function	Package	Description of function
	1	pd.DataFrame()	pandas	Creates a structured DataFrame from the city and population data.
	2	px.bar()	plotly.express	Creates an interactive bar chart from the DataFrame.
	3	update_traces()	plotly	Updates trace properties such as the position of value labels on the bars.
	4	update_layout()	plotly	Updates chart layout properties including axis titles.
	5	show()	plotly	Displays the interactive Plotly figure in the Jupyter Notebook.
	6	sort_values()	pandas	Sorts the DataFrame based on population values in ascending or descending order.
	7	idxmax()	pandas	Returns the index of the row containing

Student Reg. No: 26MML0031 Name: Hasnain Makada

Faculty: Dr Rajasekhara Babu M, Professor, School of CSE, Vellore Institute of Technology (VIT), Vellore

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				the maximum population value.
	8	idxmin()	pandas	Returns the index of the row containing the minimum population value.
	9	mean()	pandas	Calculates the average population of all cities.
	10	round()	Python	Rounds the calculated average population to the required decimal places.
	11	write_html()	plotly	Saves the interactive Plotly chart as an HTML file.
	12	to_csv()	pandas	Saves the DataFrame as a CSV file without the index column.
Steps/ Process/ Procedure/ Algorithm/ Flowchart/ Model	1. Import the Pandas library and Plotly Express for data handling and interactive visualization. 2. Create a dictionary containing the names of six Indian cities and their population values in lakhs. 3. Convert the dictionary into a Pandas DataFrame and display the dataset. 4. Check the dataset shape and print the city and population lists to verify the input data. 5. Create an interactive bar chart using Plotly Express with City on the X-axis and Population (Lakhs) on the Y-axis. 6. Display population values above the bars and provide appropriate chart and axis titles. 7. Sort the DataFrame in descending order of population using <code>sort_values()</code>. 8. Create and display a second interactive bar chart using the sorted population data. 9. Use <code>idxmax()</code> and <code>idxmin()</code> to identify the cities with the highest and lowest populations. 10. Calculate the average population using <code>mean()</code> and round the result to two decimal places. 11. Display the final population analysis summary in the notebook. 12. Save the interactive Plotly chart as <code>city_population_chart.html</code> and save the dataset as <code>city_population.csv</code>. Flow: START → Import Libraries → Create Dataset → DataFrame → Verify Data → Interactive Bar Chart → Sort Data → Sorted Chart → Population Statistics → Save HTML/CSV → STOP			
Program	import pandas as pd			

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```
import plotly.express as px

data = {
    "City": [
        "Mumbai",
        "Delhi",
        "Bengaluru",
        "Hyderabad",
        "Chennai",
        "Kolkata"
    ],
    "Population (Lakhs)": [
        204,
        167,
        136,
        108,
        115,
        149
    ]
}

df = pd.DataFrame(data)
df

print("Dataset Shape:", df.shape)
print("\nCities:")
print(df["City"].tolist())
print("\nPopulation:")
print(df["Population (Lakhs)"].tolist())

fig = px.bar(
    df,
    x="City",
    y="Population (Lakhs)",
    title="Population of Major Indian Cities",
    text="Population (Lakhs)"
)
fig.update_traces(textposition="outside")
fig.update_layout(
    xaxis_title="City",
    yaxis_title="Population (Lakhs)"
)
fig.show()

sorted_df = df.sort_values(
    by="Population (Lakhs)",
```

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```
        ascending=False
    )
    sorted_df

    fig = px.bar(
        sorted_df,
        x="City",
        y="Population (Lakhs)",
        title="City-wise Population Analysis",
        text="Population (Lakhs)"
    )
    fig.update_traces(textposition="outside")
    fig.update_layout(
        xaxis_title="City",
        yaxis_title="Population (Lakhs)"
    )
    fig.show()

    highest = df.loc[
        df["Population (Lakhs)"].idxmax()
    ]
    print("City with Highest Population:")
    print(highest)

    lowest = df.loc[
        df["Population (Lakhs)"].idxmin()
    ]
    print(
        "Lowest Population:",
        lowest["City"],
        "-",
        lowest["Population (Lakhs)"],
        "Lakhs"
    )

    average_population = df["Population (Lakhs)"].mean()
    print(
        "Average Population:",
        round(average_population, 2),
        "Lakhs"
    )

    print("CITY POPULATION ANALYSIS")
    print("-" * 35)
    print("Total Cities:", len(df))
    print(
```

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```
"Highest Population:",
df.loc[df["Population (Lakhs)"].idxmax(), "City"]
)
print(
    "Lowest Population:",
    df.loc[df["Population (Lakhs)"].idxmin(), "City"]
)
print(
    "Average Population:",
    round(df["Population (Lakhs)"].mean(), 2),
    "Lakhs"
)

fig.write_html(
    "city_population_chart.html"
)
print("Interactive chart saved successfully.")

df.to_csv(
    "city_population.csv",
    index=False
)
print("Dataset saved successfully.")
```

**Output
Snapshots
With
Explanation/
Observations**

The following output snapshots are taken from the completed Jupyter Notebook. Each image is inserted at full content width so that the dataset, charts, code cells, and output values remain readable and are not cut off.

```
[5]:      City  Population (Lakhs)
0    Mumbai             204
1      Delhi             167
2  Bengaluru             136
3   Hyderabad             108
4     Chennai             115
5     Kolkata             149

[6]: print("Dataset Shape:", df.shape)

print("\nCities:")
print(df["City"].tolist())

print("\nPopulation:")
print(df["Population (Lakhs)"].tolist())

Dataset Shape: (6, 2)

Cities:
['Mumbai', 'Delhi', 'Bengaluru', 'Hyderabad', 'Chennai', 'Kolkata']

Population:
[204, 167, 136, 108, 115, 149]

[8]: fig = px.bar(
      df,
```

Fig. 1.1: Population Dataset and Dataset Structure

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Explanation/Observation: The DataFrame verifies 6 cities and their corresponding population values.



Fig. 1.2: Interactive Bar Chart of Major Indian Cities

Explanation/Observation: Plotly Express generates an interactive bar chart for the original city order. The population values are displayed above the bars, with Mumbai showing the highest value of 204 lakhs in the given dataset.

```
[9]: sorted_df = df.sort_values(
    by="Population (Lakhs)",
    ascending=False
)

sorted_df
```

	City	Population (Lakhs)
0	Mumbai	204
1	Delhi	167
5	Kolkata	149
2	Bengaluru	136
4	Chennai	115
3	Hyderabad	108

Fig. 1.3: Population Data Sorted in Descending Order

Explanation/Observation: The DataFrame is sorted using `sort_values()` in descending order of population. The resulting order is Mumbai, Delhi, Kolkata, Bengaluru, Chennai, and Hyderabad.



Fig. 1.4: City-wise Population Analysis after Sorting

Explanation/Observation: The sorted bar chart displays the cities from

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highest to lowest population, making comparison and ranking easier.

```
[11]: highest = df.loc[
      df["Population (Lakhs)"].idxmax()
    ]

    print("City with Highest Population:")
    print(highest)
```

```
City with Highest Population:
City                Mumbai
Population (Lakhs)    204
Name: 0, dtype: object
```

```
[12]: lowest = df.loc[
      df["Population (Lakhs)"].idxmin()
    ]

    print(
        "Lowest Population:",
        lowest["City"],
        "-",
        lowest["Population (Lakhs)"],
        "Lakhs"
    )
```

```
Lowest Population: Hyderabad - 108 Lakhs
```

```
[13]: average_population = df["Population (Lakhs)"].mean()

    print(
        "Average Population:",
        round(average_population, 2),
        "Lakhs"
    )
```

```
Average Population: 146.5 Lakhs
```

Fig. 1.5: Highest, Lowest, and Average Population Analysis

Explanation/Observation: The program identifies Mumbai as the city with the highest population (204 lakhs), Hyderabad as the city with the lowest population (108 lakhs), and calculates the average population as 146.5 lakhs.

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```
[14]: print("CITY POPULATION ANALYSIS")
      print("-" * 35)

      print("Total Cities:", len(df))

      print(
          "Highest Population:",
          df.loc[df["Population (Lakhs)"].idxmax(), "City"]
      )

      print(
          "Lowest Population:",
          df.loc[df["Population (Lakhs)"].idxmin(), "City"]
      )

      print(
          "Average Population:",
          round(df["Population (Lakhs)"].mean(), 2),
          "Lakhs"
      )

CITY POPULATION ANALYSIS
-----
Total Cities: 6
Highest Population: Mumbai
Lowest Population: Hyderabad
Average Population: 146.5 Lakhs

[15]: fig.write_html(
      "city_population_chart.html"
      )

      print("Interactive chart saved successfully.")

Interactive chart saved successfully.

[16]: df.to_csv(
      "city_population.csv",
      index=False
      )

      print("Dataset saved successfully.")

Dataset saved successfully.

[ ]:
```

Fig. 1.6: Final Analysis Summary and File Saving Output

Explanation/Observation: The final summary confirms 6 cities were analyzed, with Mumbai highest and Hyderabad lowest. The interactive chart and CSV dataset are saved successfully.

Conclusions	The experiment was successfully completed using Pandas and Plotly. The population data was organized in a DataFrame, visualized with interactive bar charts, and sorted for comparison. Mumbai had the highest population, Hyderabad the lowest, and the average population was 146.5 lakhs. The interactive chart was saved as HTML and the dataset as CSV.
References	1. Michael Freeman and Joel Ross, Programming Skills for Data Science, Addison-Wesley, 2018. 2. Python Program (GitHub link): https://github.com/hasnainmakada-99/Data-Science-Codes/blob/main/Week%206/Week-6-PP.ipynb 3. Class lecture link: Open Lecture File

Student Reg. No: 26MML0031 Name: Hasnain Makada

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